

TE Curve Verification Pipeline Directional Model Comparison

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report is the canonical TE Curve Verification Pipeline offline comparison between RCIM Model-Bank Reproduction , recovered original, retuned paper-reference model banks, and repository-owned Wave 1 and Wave 2.1 model candidates. It starts from the current direction-aware comparison matrix.

Dataset And Split

- dataset config: config/datasets/transmission_error_dataset.yaml ;
- dataset root: data\simplified_dataset ;
- comparison mode: full_directional_candidate_matrix ;
- candidate count: 231 ;
- held-out curve count before candidate filtering: 194 ;
- percentage-error denominator: peak_to_peak_truth ;
- Fw candidates are evaluated only on forward curves;
- Bw candidates are evaluated only on backward curves;
- global candidates are evaluated on both directions and reported with direction-separated metrics.

Candidate Inventory

Candidate	Family	Source	Kind	Surface	Valid Directions
SVM19_Fw	SVM	rcim_track1	track1_reference_bank	Fw	forward
SVM19_Bw	SVM	rcim_track1	track1_reference_bank	Bw	backward
MLP19_Fw	MLP	rcim_track1	track1_reference_bank	Fw	forward
MLP19_Bw	MLP	rcim_track1	track1_reference_bank	Bw	backward
RF19_Fw	RF	rcim_track1	track1_reference_bank	Fw	forward
RF19_Bw	RF	rcim_track1	track1_reference_bank	Bw	backward
DT19_Fw	DT	rcim_track1	track1_reference_bank	Fw	forward
DT19_Bw	DT	rcim_track1	track1_reference_bank	Bw	backward

Candidate	Family	Source	Kind	Surface	Valid Directions
ET19_Fw	ET	rcim_track1	track1_reference_bank	Fw	forward
ET19_Bw	ET	rcim_track1	track1_reference_bank	Bw	backward
ERT19_Fw	ERT	rcim_track1	track1_reference_bank	Fw	forward
ERT19_Bw	ERT	rcim_track1	track1_reference_bank	Bw	backward
GBM19_Fw	GBM	rcim_track1	track1_reference_bank	Fw	forward
GBM19_Bw	GBM	rcim_track1	track1_reference_bank	Bw	backward
HGBM19_Fw	HGBM	rcim_track1	track1_reference_bank	Fw	forward
HGBM19_Bw	HGBM	rcim_track1	track1_reference_bank	Bw	backward
XGBM19_Fw	XGBM	rcim_track1	track1_reference_bank	Fw	forward
XGBM19_Bw	XGBM	rcim_track1	track1_reference_bank	Bw	backward
LGBM19_Fw	LGBM	rcim_track1	track1_reference_bank	Fw	forward
LGBM19_Bw	LGBM	rcim_track1	track1_reference_bank	Bw	backward
ELM19_Fw	ELM	rcim_track1	track1_reference_bank	Fw	forward
ELM19_Bw	ELM	rcim_track1	track1_reference_bank	Bw	backward
rcim_original_SVM19_Fw	SVM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_MLP19_Fw	MLP	rcim_original	track1_reference_bank	Fw	forward
rcim_original_RF19_Fw	RF	rcim_original	track1_reference_bank	Fw	forward
rcim_original_DT19_Fw	DT	rcim_original	track1_reference_bank	Fw	forward
rcim_original_ET19_Fw	ET	rcim_original	track1_reference_bank	Fw	forward
rcim_original_ERT19_Fw	ERT	rcim_original	track1_reference_bank	Fw	forward
rcim_original_GBM19_Fw	GBM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_HGBM19_Fw	HGBM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_XGBM19_Fw	XGBM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_LGBM19_Fw	LGBM	rcim_original	track1_reference_bank	Fw	forward
rcim_original_ELM19_Fw	ELM	rcim_original	track1_reference_bank	Fw	forward
rcim_retuned_SVM19_Fw	SVM	rcim_retuned	track1_reference_bank	Fw	forward

Candidate	Family	Source	Kind	Surface	Valid Directions
rcim_retuned_MLP19_Fw	MLP	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_RF19_Fw	RF	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_DT19_Fw	DT	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_ET19_Fw	ET	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_ERT19_Fw	ERT	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_GBM19_Fw	GBM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_HGBM19_Fw	HGBM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_XGBM19_Fw	XGBM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_LGBM19_Fw	LGBM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_ELM19_Fw	ELM	rcim_retuned	track1_reference_bank	Fw	forward
rcim_retuned_SVM19_Bw	SVM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_MLP19_Bw	MLP	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_RF19_Bw	RF	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_DT19_Bw	DT	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_ET19_Bw	ET	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_ERT19_Bw	ERT	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_GBM19_Bw	GBM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_HGBM19_Bw	HGBM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_XGBM19_Bw	XGBM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_LGBM19_Bw	LGBM	rcim_retuned	track1_reference_bank	Bw	backward
rcim_retuned_ELM19_Bw	ELM	rcim_retuned	track1_reference_bank	Bw	backward
temporal_convolution_global	temporal_convolution	wave2_temporal_entry_registry	wave1_registry_model	global	forward, backward
temporal_convolution_Fw	temporal_convolution	wave2_temporal_entry_registry	wave1_registry_model	Fw	forward
temporal_convolution_Bw	temporal_convolution	wave2_temporal_entry_registry	wave1_registry_model	Bw	backward
gru_sequence_global	gru_sequence	wave2_temporal_entry_registry	wave1_registry_model	global	forward, backward
gru_sequence_Fw	gru_sequence	wave2_temporal_entry_registry	wave1_registry_model	Fw	forward

Candidate	Family	Source	Kind	Surface	Valid Directions
gru_sequence_Bw	gru_sequence	wave2_temporal_entry_registry	wave1_registry_model	Bw	backward
lstm_sequence_global	lstm_sequence	wave2_temporal_entry_registry	wave1_registry_model	global	forward, backward
lstm_sequence_Fw	lstm_sequence	wave2_temporal_entry_registry	wave1_registry_model	Fw	forward
lstm_sequence_Bw	lstm_sequence	wave2_temporal_entry_registry	wave1_registry_model	Bw	backward
feedforward_global	feedforward	wave1	wave1_exported_model	global	forward, backward
feedforward_Fw	feedforward	wave1	wave1_exported_model	Fw	forward
feedforward_Bw	feedforward	wave1	wave1_exported_model	Bw	backward
harmonic_regression_global	harmonic_regression	wave1	wave1_exported_model	global	forward, backward
harmonic_regression_Fw	harmonic_regression	wave1	wave1_exported_model	Fw	forward
harmonic_regression_Bw	harmonic_regression	wave1	wave1_exported_model	Bw	backward
periodic_mlp_global	periodic_mlp	wave1	wave1_exported_model	global	forward, backward
periodic_mlp_Fw	periodic_mlp	wave1	wave1_exported_model	Fw	forward
periodic_mlp_Bw	periodic_mlp	wave1	wave1_exported_model	Bw	backward
residual_harmonic_mlp_global	residual_harmonic_mlp	wave1	wave1_exported_model	global	forward, backward
residual_harmonic_mlp_Fw	residual_harmonic_mlp	wave1	wave1_exported_model	Fw	forward
residual_harmonic_mlp_Bw	residual_harmonic_mlp	wave1	wave1_exported_model	Bw	backward
tree_global	tree	wave1	wave1_exported_model	global	forward, backward
tree_Fw	tree	wave1	wave1_exported_model	Fw	forward
tree_Bw	tree	wave1	wave1_exported_model	Bw	backward
paper_original_best_Fw	best_composite	rcim_original	composite_reference_bank	Fw	forward
paper_retuned_best_Fw	best_composite	rcim_retuned	composite_reference_bank	Fw	forward
track1_best_Fw	best_composite	rcim_track1	composite_reference_bank	Fw	forward
paper_retuned_best_Bw	best_composite	rcim_retuned	composite_reference_bank	Bw	backward
track1_best_Bw	best_composite	rcim_track1	composite_reference_bank	Bw	backward
periodic_temporal_convolution_global	periodic_temporal_convolution	wave2_temporal_entry_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
periodic_temporal_convolution_Fw	periodic_temporal_convolution	wave2_temporal_entropy_registry	wave1_registry_model	Fw	forward
periodic_temporal_convolution_Bw	periodic_temporal_convolution	wave2_temporal_entropy_registry	wave1_registry_model	Bw	backward
periodic_gru_sequence_global	periodic_gru_sequence	wave2_temporal_entropy_registry	wave1_registry_model	global	forward, backward
periodic_gru_sequence_Fw	periodic_gru_sequence	wave2_temporal_entropy_registry	wave1_registry_model	Fw	forward
periodic_gru_sequence_Bw	periodic_gru_sequence	wave2_temporal_entropy_registry	wave1_registry_model	Bw	backward
periodic_lstm_sequence_global	periodic_lstm_sequence	wave2_temporal_entropy_registry	wave1_registry_model	global	forward, backward
periodic_lstm_sequence_Fw	periodic_lstm_sequence	wave2_temporal_entropy_registry	wave1_registry_model	Fw	forward
periodic_lstm_sequence_Bw	periodic_lstm_sequence	wave2_temporal_entropy_registry	wave1_registry_model	Bw	backward
residual_harmonic_gru_sequence_sparse_rcim_global	residual_harmonic_gru_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_gru_sequence_sparse_rcim_Fw	residual_harmonic_gru_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_gru_sequence_sparse_rcim_Bw	residual_harmonic_gru_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_gru_sequence_dense240_global	residual_harmonic_gru_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_gru_sequence_dense240_Fw	residual_harmonic_gru_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_gru_sequence_dense240_Bw	residual_harmonic_gru_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_gru_sequence_dense360_global	residual_harmonic_gru_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_gru_sequence_dense360_Fw	residual_harmonic_gru_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_gru_sequence_dense360_Bw	residual_harmonic_gru_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_lstm_sequence_sparse_rcim_global	residual_harmonic_lstm_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
residual_harmonic_lstm_sequence_sparse_rcim_Fw	residual_harmonic_lstm_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_lstm_sequence_sparse_rcim_Bw	residual_harmonic_lstm_sequence_sparse_rcim	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_lstm_sequence_dense240_global	residual_harmonic_lstm_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_lstm_sequence_dense240_Fw	residual_harmonic_lstm_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_lstm_sequence_dense240_Bw	residual_harmonic_lstm_sequence_dense240	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
residual_harmonic_lstm_sequence_dense360_global	residual_harmonic_lstm_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	global	forward, backward
residual_harmonic_lstm_sequence_dense360_Fw	residual_harmonic_lstm_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Fw	forward
residual_harmonic_lstm_sequence_dense360_Bw	residual_harmonic_lstm_sequence_dense360	wave2c_residual_harmonic_temporal_registry	wave1_registry_model	Bw	backward
polished_rcim_model_bank_reproduction_SVM19_Bw	SVM	polished_rcim_model_bank_reproduction	track1_reference_bank	Bw	backward
polished_rcim_model_bank_reproduction_MLP19_Bw	MLP	polished_rcim_model_bank_reproduction	track1_reference_bank	Bw	backward
polished_rcim_model_bank_reproduction_RF19_Bw	RF	polished_rcim_model_bank_reproduction	track1_reference_bank	Bw	backward
polished_rcim_model_bank_reproduction_DT19_Bw	DT	polished_rcim_model_bank_reproduction	track1_reference_bank	Bw	backward
polished_rcim_model_bank_reproduction_ET19_Bw	ET	polished_rcim_model_bank_reproduction	track1_reference_bank	Bw	backward
polished_rcim_model_bank_reproduction_ERT19_Bw	ERT	polished_rcim_model_bank_reproduction	track1_reference_bank	Bw	backward
polished_rcim_model_bank_reproduction_GBM19_Bw	GBM	polished_rcim_model_bank_reproduction	track1_reference_bank	Bw	backward
polished_rcim_model_bank_reproduction_HGBM19_Bw	HGBM	polished_rcim_model_bank_reproduction	track1_reference_bank	Bw	backward

Candidate	Family	Source	Kind	Surface	Valid Directions
reproduction_HGBM19_Bw		on			
polished_rcim_model_bank_reproduction_XGBM19_Bw	XGBM	polished_rcim_model_bank_reproduction	track1_reference_bank	Bw	backward
polished_rcim_model_bank_reproduction_LGBM19_Bw	LGBM	polished_rcim_model_bank_reproduction	track1_reference_bank	Bw	backward
sequential_residual_offset_probe_global	sequential_residual_offset_probe	track2f_offset_aware_probe_registry	wave1_registry_model	global	forward, backward
sequential_residual_offset_probe_Bw	sequential_residual_offset_probe	track2f_offset_aware_probe_registry	wave1_registry_model	Bw	backward
track2f_bis_clean_sequential_residual_offset_global	track2f_bis_clean_sequential_residual_offset	track2f_bis_harmonic_offset_probe_registry	wave1_registry_model	global	forward, backward
track2f_bis_clean_sequential_residual_offset_Bw	track2f_bis_clean_sequential_residual_offset	track2f_bis_harmonic_offset_probe_registry	wave1_registry_model	Bw	backward
track2f_bis_harmonic_residual_offset_global	track2f_bis_harmonic_residual_offset	track2f_bis_harmonic_offset_probe_registry	wave1_registry_model	global	forward, backward
track2f_bis_harmonic_residual_offset_Bw	track2f_bis_harmonic_residual_offset	track2f_bis_harmonic_offset_probe_registry	wave1_registry_model	Bw	backward
track2g_curve_aware_pointwise_control_global	track2g_curve_aware_pointwise_control	track2g_curve_aware_training_registry	wave1_registry_model	global	forward, backward
track2g_curve_aware_pointwise_control_Bw	track2g_curve_aware_pointwise_control	track2g_curve_aware_training_registry	wave1_registry_model	Bw	backward
track2g_curve_aware_raw_centered_shape_global	track2g_curve_aware_raw_centered_shape	track2g_curve_aware_training_registry	wave1_registry_model	global	forward, backward
track2g_curve_aware_raw_centered_shape_Bw	track2g_curve_aware_raw_centered_shape	track2g_curve_aware_training_registry	wave1_registry_model	Bw	backward
track2g_curve_aware_raw_offset_global	track2g_curve_aware_raw_offset	track2g_curve_aware_training_registry	wave1_registry_model	global	forward, backward
track2g_curve_aware_raw_offset_Bw	track2g_curve_aware_raw_offset	track2g_curve_aware_training_registry	wave1_registry_model	Bw	backward
track2g_curve_aware_full_curve_composite_global	track2g_curve_aware_full_curve_composite	track2g_curve_aware_training_registry	wave1_registry_model	global	forward, backward
track2g_curve_aware_full_curve_composite_Bw	track2g_curve_aware_full_curve_composite	track2g_curve_aware_training_registry	wave1_registry_model	Bw	backward
track2h_mae_robust_global	track2h_mae_robust	track2h_dispersion_aware_modeling_registry	wave1_registry_model	global	forward, backward
track2h_mae_robust_Bw	track2h_mae_robust	track2h_dispersion_aware_modeling_registry	wave1_registry_model	Bw	backward

Candidate	Family	Source	Kind	Surface	Valid Directions
egistry					
track2h_smooth_l1_robust_global	track2h_smooth_l1_robust	track2h_dispersion_aware_modeling_registry	wave1_registry_model	global	forward, backward
track2h_smooth_l1_robust_Bw	track2h_smooth_l1_robust	track2h_dispersion_aware_modeling_registry	wave1_registry_model	Bw	backward
track2h_log_cosh_robust_global	track2h_log_cosh_robust	track2h_dispersion_aware_modeling_registry	wave1_registry_model	global	forward, backward
track2h_log_cosh_robust_Bw	track2h_log_cosh_robust	track2h_dispersion_aware_modeling_registry	wave1_registry_model	Bw	backward
track2h_quantile_p10_p50_p90_global	track2h_quantile_p10_p50_p90	track2h_quantile_probabilistic_registry	wave1_registry_model	global	forward, backward
track2h_quantile_p10_p50_p90_Bw	track2h_quantile_p10_p50_p90	track2h_quantile_probabilistic_registry	wave1_registry_model	Bw	backward
track2h_gaussian_nll_global	track2h_gaussian_nll	track2h_quantile_probabilistic_registry	wave1_registry_model	global	forward, backward
track2h_gaussian_nll_Bw	track2h_gaussian_nll	track2h_quantile_probabilistic_registry	wave1_registry_model	Bw	backward
track2h_mdn_k2_global	track2h_mdn_k2	track2h_mixture_density_heads_registry	wave1_registry_model	global	forward, backward
track2h_mdn_k2_Bw	track2h_mdn_k2	track2h_mixture_density_heads_registry	wave1_registry_model	Bw	backward
track2h_mdn_k3_global	track2h_mdn_k3	track2h_mixture_density_heads_registry	wave1_registry_model	global	forward, backward
track2h_mdn_k3_Bw	track2h_mdn_k3	track2h_mixture_density_heads_registry	wave1_registry_model	Bw	backward
track2h_l_gru_offset_residual_global	track2h_l_gru_offset_residual	track2h_latent_state_hysteresis_registry	wave1_registry_model	global	forward, backward
track2h_l_gru_offset_residual_Bw	track2h_l_gru_offset_residual	track2h_latent_state_hysteresis_registry	wave1_registry_model	Bw	backward
track2h_l_causal_tcn_offset_residual_global	track2h_l_causal_tcn_offset_residual	track2h_latent_state_hysteresis_registry	wave1_registry_model	global	forward, backward
track2h_l_causal_tcn_offset_residual_Bw	track2h_l_causal_tcn_offset_residual	track2h_latent_state_hysteresis_registry	wave1_registry_model	Bw	backward
wave3_harmonic_prior_residual_pointwise_control_global	wave3_harmonic_prior_residual_pointwise_control	wave3_harmonic_prior_residual_registry	wave1_registry_model	global	forward, backward
wave3_harmonic_prior_residual_pointwise_control_Bw	wave3_harmonic_prior_residual_pointwise_control	wave3_harmonic_prior_residual_registry	wave1_registry_model	Bw	backward
wave3_harmonic_prior_residual_	wave3_harmonic_prior_residual_smooth	wave3_harmonic_prior_residual_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
smooth_l1_structured_global	h_l1_structured	try			
wave3_harmonic_prior_residual_smooth_l1_structured_Bw	wave3_harmonic_prior_residual_smooth_l1_structured	wave3_harmonic_prior_residual_registry	wave1_registry_model	Bw	backward
wave52b_offset_centered_shape_harmonic_global	wave52b_offset_harmonic_guided	wave52b_offset_harmonic_guided_registry	wave1_registry_model	global	forward, backward
wave52b_offset_centered_shape_harmonic_Bw	wave52b_offset_harmonic_guided	wave52b_offset_harmonic_guided_registry	wave1_registry_model	Bw	backward
polished_feedforward_global	feedforward	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_feedforward_Bw	feedforward	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_harmonic_regression_global	harmonic_regression	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_harmonic_regression_Bw	harmonic_regression	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_periodic_mlp_global	periodic_mlp	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_mlp_Bw	periodic_mlp	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_residual_harmonic_mlp_global	residual_harmonic_mlp	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_mlp_Bw	residual_harmonic_mlp	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_tree_global	tree	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_tree_Bw	tree	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_periodic_mlp_harmonic_global	periodic_mlp_harmonic	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_mlp_harmonic_Bw	periodic_mlp_harmonic	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_temporal_convolution_global	temporal_convolution	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_temporal_convolution_Bw	temporal_convolution	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_gru_sequence_global	gru_sequence	polished_model_development_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
polished_gru_sequence_Bw	gru_sequence	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_lstm_sequence_global	lstm_sequence	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_lstm_sequence_Bw	lstm_sequence	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_periodic_temporal_convolution_global	periodic_temporal_convolution	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_temporal_convolution_Bw	periodic_temporal_convolution	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_periodic_gru_sequence_global	periodic_gru_sequence	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_gru_sequence_Bw	periodic_gru_sequence	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_periodic_lstm_sequence_global	periodic_lstm_sequence	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_periodic_lstm_sequence_Bw	periodic_lstm_sequence	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_residual_harmonic_gru_sequence_sparse_rcim_global	residual_harmonic_gru_sequence_sparse_rcim	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_gru_sequence_sparse_rcim_Bw	residual_harmonic_gru_sequence_sparse_rcim	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_residual_harmonic_gru_sequence_dense240_global	residual_harmonic_gru_sequence_dense240	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_gru_sequence_dense240_Bw	residual_harmonic_gru_sequence_dense240	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_residual_harmonic_gru_sequence_dense360_global	residual_harmonic_gru_sequence_dense360	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_gru_sequence_dense360_Bw	residual_harmonic_gru_sequence_dense360	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	residual_harmonic_lstm_sequence_sparse_rcim	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_lstm_sequence_sparse_rcim_Bw	residual_harmonic_lstm_sequence_sparse_rcim	polished_model_development_registry	wave1_registry_model	Bw	backward

Candidate	Family	Source	Kind	Surface	Valid Directions
lstm_sequence_sparse_rcim_Bw					
polished_residual_harmonic_lstm_sequence_dense240_global	residual_harmonic_lstm_sequence_dense240	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_lstm_sequence_dense240_Bw	residual_harmonic_lstm_sequence_dense240	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_residual_harmonic_lstm_sequence_dense360_global	residual_harmonic_lstm_sequence_dense360	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_residual_harmonic_lstm_sequence_dense360_Bw	residual_harmonic_lstm_sequence_dense360	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_wave3_1_sequential_residual_offset_probe_global	wave3_1_sequential_residual_offset_probe	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_1_sequential_residual_offset_probe_Bw	wave3_1_sequential_residual_offset_probe	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_wave3_2_clean_sequential_residual_offset_global	wave3_2_clean_sequential_residual_offset	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_2_clean_sequential_residual_offset_Bw	wave3_2_clean_sequential_residual_offset	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_wave3_2_harmonic_residual_offset_global	wave3_2_harmonic_residual_offset	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_2_harmonic_residual_offset_Bw	wave3_2_harmonic_residual_offset	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_wave3_3_curve_aware_pointwise_control_global	wave3_3_curve_aware_pointwise_control	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_3_curve_aware_pointwise_control_Bw	wave3_3_curve_aware_pointwise_control	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_wave3_3_raw_centered_shape_curve_aware_global	wave3_3_raw_centered_shape_curve_aware	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_3_raw_centered_shape_curve_aware_Bw	wave3_3_raw_centered_shape_curve_aware	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_wave3_3_raw_offset_curve_aware_global	wave3_3_raw_offset_curve_aware	polished_model_development_registry	wave1_registry_model	global	forward, backward

Candidate	Family	Source	Kind	Surface	Valid Directions
polished_wave3_3_raw_offset_curve_aware_Bw	wave3_3_raw_offset_curve_aware	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_wave3_3_full_curve_composite_global	wave3_3_full_curve_composite	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave3_3_full_curve_composite_Bw	wave3_3_full_curve_composite	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_wave4_1_mae_robust_loss_global	wave4_1_mae_robust_loss	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_1_mae_robust_loss_Bw	wave4_1_mae_robust_loss	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_wave4_1_smooth_l1_robust_loss_global	wave4_1_smooth_l1_robust_loss	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_1_smooth_l1_robust_loss_Bw	wave4_1_smooth_l1_robust_loss	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_wave4_1_log_cosh_robust_loss_global	wave4_1_log_cosh_robust_loss	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_1_log_cosh_robust_loss_Bw	wave4_1_log_cosh_robust_loss	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_wave4_2_quantile_p10_p50_p90_global	wave4_2_quantile_p10_p50_p90	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_2_quantile_p10_p50_p90_Bw	wave4_2_quantile_p10_p50_p90	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_wave4_2_gaussian_nll_global	wave4_2_gaussian_nll	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_2_gaussian_nll_Bw	wave4_2_gaussian_nll	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_wave4_3_mixture_density_k2_global	wave4_3_mixture_density_k2	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_3_mixture_density_k2_Bw	wave4_3_mixture_density_k2	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_wave4_3_mixture_density_k3_global	wave4_3_mixture_density_k3	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_3_mixture_density_k3_Bw	wave4_3_mixture_density_k3	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_wave4_4_gru_latent_offset_residual_global	wave4_4_gru_latent_offset_residual	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_4_gru_latent_offset_residual_Bw	wave4_4_gru_latent_offset_residual	polished_model_development_registry	wave1_registry_model	Bw	backward

Candidate	Family	Source	Kind	Surface	Valid Directions
polished_wave4_4_causal_tcn_latent_offset_residual_global	wave4_4_causal_tcn_latent_offset_residual	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave4_4_causal_tcn_latent_offset_residual_Bw	wave4_4_causal_tcn_latent_offset_residual	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_wave5_1_harmonic_prior_pointwise_control_global	wave5_1_harmonic_prior_pointwise_control	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave5_1_harmonic_prior_pointwise_control_Bw	wave5_1_harmonic_prior_pointwise_control	polished_model_development_registry	wave1_registry_model	Bw	backward
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	wave5_1_harmonic_prior_smooth_l1_structured	polished_model_development_registry	wave1_registry_model	global	forward, backward
polished_wave5_1_harmonic_prior_smooth_l1_structured_Bw	wave5_1_harmonic_prior_smooth_l1_structured	polished_model_development_registry	wave1_registry_model	Bw	backward

Best Composite Reference Models

These candidates combine the approved best harmonic-wise cells into one TE Curve Verification Pipeline curve-reconstruction candidate. They are also repeated inside the source-group tables below, but this section keeps the composed models explicit.

Candidate	Source	Surface	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
paper_original_best_Fw	rcim_original	Fw	forward	0.002769	0.002951	6.250	13.827
paper_retuned_best_Fw	rcim_retuned	Fw	forward	0.001839	0.002041	4.109	9.866
track1_best_Fw	rcim_track1	Fw	forward	0.003014	0.003204	6.819	11.638
paper_retuned_best_Bw	rcim_retuned	Bw	backward	0.003675	0.004284	7.572	15.645
track1_best_Bw	rcim_track1	Bw	backward	0.005027	0.005212	11.860	48.106

Forward Comparison

Original Forward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
rcim_original_ERT19_Fw	0.001471	0.001677	3.253	6.145
rcim_original_RF19_Fw	0.001767	0.001971	3.940	6.872
rcim_original_LGBM19_Fw	0.001801	0.002004	4.017	10.054
rcim_original_DT19_Fw	0.001919	0.002114	4.306	9.063
rcim_original_GBM19_Fw	0.001921	0.002122	4.312	8.193
rcim_original_HGBM19_Fw	0.002011	0.002217	4.493	10.617
rcim_original_ET19_Fw	0.002232	0.002432	4.985	11.357
rcim_original_XGBM19_Fw	0.002594	0.002814	5.805	10.574
paper_original_best_Fw	0.002769	0.002951	6.250	13.827
rcim_original_SVM19_Fw	0.003052	0.003324	6.767	13.827
rcim_original_ELM19_Fw	0.005423	0.005731	12.130	28.721
rcim_original_MLP19_Fw	0.018754	0.022589	42.943	107.515

Retuned Forward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
rcim_retuned_GBM19_Fw	0.001089	0.001299	2.372	4.912
rcim_retuned_RF19_Fw	0.001487	0.001699	3.292	5.998
rcim_retuned_ERT19_Fw	0.001807	0.002010	4.039	7.599
paper_retuned_best_Fw	0.001839	0.002041	4.109	9.866
rcim_retuned_HGBM19_Fw	0.001851	0.002056	4.126	9.401
rcim_retuned_LGBM19_Fw	0.001851	0.002055	4.135	9.866
rcim_retuned_DT19_Fw	0.001919	0.002114	4.306	9.063
rcim_retuned_ET19_Fw	0.002001	0.002196	4.426	9.533
rcim_retuned_XGBM19_Fw	0.002054	0.002264	4.588	10.488
rcim_retuned_SVM19_Fw	0.003167	0.003428	7.035	12.971
rcim_retuned_ELM19_Fw	0.007182	0.007463	16.181	40.024
rcim_retuned_MLP19_Fw	0.016647	0.020154	38.510	86.197

RCIM Model-Bank Reproduction Forward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
RF19_Fw	0.002164	0.002371	4.841	11.061
ERT19_Fw	0.002295	0.002491	5.163	13.141
GBM19_Fw	0.002779	0.002981	6.238	12.995
track1_best_Fw	0.003014	0.003204	6.819	11.638
DT19_Fw	0.003122	0.003313	7.011	13.928
SVM19_Fw	0.003236	0.003487	7.185	11.841
HGBM19_Fw	0.003251	0.003464	7.315	13.802
ET19_Fw	0.003267	0.003467	7.339	14.670
XGBM19_Fw	0.004190	0.004396	9.407	22.660
LGBM19_Fw	0.006812	0.007009	15.415	30.398
ELM19_Fw	0.007281	0.007573	16.352	36.326
MLP19_Fw	0.038690	0.047157	89.567	201.437

Wave 1 Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
tree_global	0.002998	0.003364	6.590	11.601
tree_Fw	0.003053	0.003395	6.731	11.995
periodic_mlp_Fw	0.003366	0.003794	7.467	12.904
periodic_mlp_global	0.003380	0.003791	7.493	14.349
harmonic_regression_Fw	0.003394	0.003807	7.534	11.391
residual_harmonic_mlp_Fw	0.003434	0.003876	7.618	13.617
feedforward_Fw	0.003466	0.003904	7.685	13.132
feedforward_global	0.003550	0.003979	7.916	16.195
residual_harmonic_mlp_global	0.003649	0.004051	8.123	14.713
harmonic_regression_global	0.018314	0.018547	41.749	79.404

Wave 2.1 Temporal Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
periodic_lstm_sequence_global	0.002726	0.002959	6.142	14.092
periodic_gru_sequence_global	0.002777	0.003023	6.267	13.580
periodic_gru_sequence_Fw	0.003186	0.003438	7.077	11.974
periodic_lstm_sequence_Fw	0.003266	0.003550	7.258	11.961
gru_sequence_Fw	0.003330	0.003762	7.378	13.029
periodic_temporal_convolution_Fw	0.003335	0.003708	7.404	12.518
lstm_sequence_Fw	0.003366	0.003800	7.450	11.807
periodic_temporal_convolution_global	0.003407	0.003724	7.581	14.480
lstm_sequence_global	0.003445	0.003863	7.642	12.032
temporal_convolution_global	0.003508	0.003928	7.792	14.045
gru_sequence_global	0.003546	0.003975	7.869	14.344
temporal_convolution_Fw	0.003603	0.004031	8.028	14.674

Wave 2.3 Residual Harmonic Temporal Forward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
residual_harmonic_gru_sequence_sparse_rcim_Fw	0.003194	0.003499	7.083	13.041
residual_harmonic_lstm_sequence_sparse_rcim_Fw	0.003229	0.003533	7.164	12.682
residual_harmonic_lstm_sequence_sparse_rcim_global	0.003245	0.003562	7.192	12.555
residual_harmonic_gru_sequence_sparse_rcim_global	0.003287	0.003611	7.282	12.461
residual_harmonic_lstm_sequence_dense240_global	0.006247	0.008597	14.061	20.066
residual_harmonic_gru_sequence_dense240_global	0.006534	0.008959	14.732	20.898
residual_harmonic_gru_sequence_dense240_Fw	0.006983	0.009275	15.722	19.139
residual_harmonic_lstm_sequence_dense240_Fw	0.007042	0.009370	15.868	19.358
residual_harmonic_lstm_sequence_dense360_Fw	0.007731	0.010235	17.430	21.285
residual_harmonic_gru_sequence_dense360_global	0.007844	0.011211	17.679	22.520
residual_harmonic_gru_sequence_dense360_Fw	0.007869	0.010574	17.740	20.918
residual_harmonic_lstm_sequence_dense360_global	0.008760	0.012992	19.759	24.068

Backward Comparison

Retuned Backward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
rcim_retuned_GBM19_Bw	0.002766	0.003300	5.398	12.280
rcim_retuned_ET19_Bw	0.003441	0.004028	7.021	15.287
rcim_retuned_ERT19_Bw	0.003551	0.004161	7.269	13.187
rcim_retuned_RF19_Bw	0.003649	0.004256	7.543	15.083
paper_retuned_best_Bw	0.003675	0.004284	7.572	15.645
rcim_retuned_SVM19_Bw	0.004016	0.004599	8.813	17.215
rcim_retuned_DT19_Bw	0.004578	0.005169	9.728	19.601
rcim_retuned_HGBM19_Bw	0.004683	0.005301	9.978	17.712
rcim_retuned_LGBM19_Bw	0.008105	0.008655	18.057	35.748
rcim_retuned_ELM19_Bw	0.008917	0.009518	20.169	51.896
rcim_retuned_XGBM19_Bw	0.010679	0.011209	24.184	48.082
rcim_retuned_MLP19_Bw	0.019115	0.023025	44.141	88.991

RCIM Model-Bank Reproduction Backward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
SVM19_Bw	0.004822	0.005116	10.864	25.533
track1_best_Bw	0.005027	0.005212	11.860	48.106
LGBM19_Bw	0.005037	0.005231	11.880	48.106
GBM19_Bw	0.005180	0.005363	12.252	49.984
DT19_Bw	0.005226	0.005409	12.359	48.860
ERT19_Bw	0.005258	0.005442	12.434	51.665
RF19_Bw	0.005392	0.005584	12.731	55.740
ET19_Bw	0.006273	0.006520	14.314	48.624
HGBM19_Bw	0.006619	0.006834	15.494	53.982
XGBM19_Bw	0.007991	0.008195	18.722	59.067
ELM19_Bw	0.010071	0.010486	23.034	52.159
MLP19_Bw	0.036681	0.044921	86.544	225.831

Wave 1 Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
tree_Bw	0.003258	0.003651	7.051	14.116
tree_global	0.003290	0.003702	7.118	13.703
periodic_mlp_Bw	0.003574	0.004006	7.807	13.970
periodic_mlp_global	0.003610	0.004030	7.879	13.661
residual_harmonic_mlp_global	0.003639	0.004044	7.967	14.489
feedforward_Bw	0.003644	0.004067	7.977	14.284
residual_harmonic_mlp_Bw	0.003672	0.004089	8.046	14.973
harmonic_regression_Bw	0.003745	0.004173	8.171	15.684
feedforward_global	0.003741	0.004188	8.186	15.259
harmonic_regression_global	0.018006	0.018276	41.275	83.603

Wave 2.1 Temporal Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
periodic_gru_sequence_Bw	0.002392	0.002639	5.466	14.820
periodic_gru_sequence_global	0.002630	0.002876	6.010	12.693
periodic_lstm_sequence_Bw	0.002625	0.002877	6.013	15.382
periodic_lstm_sequence_global	0.002689	0.002956	6.098	14.674
lstm_sequence_global	0.003515	0.003944	7.666	12.502
lstm_sequence_Bw	0.003555	0.003985	7.767	14.507
gru_sequence_Bw	0.003626	0.004082	7.907	12.900
periodic_temporal_convolution_global	0.003604	0.003948	7.935	13.804
gru_sequence_global	0.003637	0.004080	7.946	14.580
periodic_temporal_convolution_Bw	0.003628	0.003987	7.979	13.839
temporal_convolution_Bw	0.003742	0.004166	8.184	13.908
temporal_convolution_global	0.003994	0.004438	8.798	18.339

Wave 2.3 Residual Harmonic Temporal Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
residual_harmonic_lstm_sequence_sparse_rcim_Bw	0.003440	0.003793	7.510	13.160
residual_harmonic_lstm_sequence_sparse_rcim_global	0.003491	0.003876	7.627	13.540
residual_harmonic_gru_sequence_sparse_rcim_Bw	0.003502	0.003857	7.654	12.829
residual_harmonic_gru_sequence_sparse_rcim_global	0.003584	0.003957	7.859	14.946
residual_harmonic_lstm_sequence_dense240_global	0.006591	0.008932	14.858	21.728
residual_harmonic_gru_sequence_dense240_global	0.006786	0.009221	15.282	20.990
residual_harmonic_lstm_sequence_dense240_Bw	0.007367	0.009945	16.660	22.200
residual_harmonic_gru_sequence_dense360_global	0.008181	0.011621	18.502	23.968
residual_harmonic_lstm_sequence_dense360_global	0.008860	0.013061	20.073	23.940
residual_harmonic_gru_sequence_dense240_Bw	0.008984	0.012987	20.358	26.370
residual_harmonic_gru_sequence_dense360_Bw	0.009370	0.013165	21.267	25.901
residual_harmonic_lstm_sequence_dense360_Bw	0.010268	0.014769	23.355	29.779

Wave 3.1 Offset-Aware Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
sequential_residual_offset_probe_Bw	0.003636	0.004065	7.952	13.795
sequential_residual_offset_probe_global	0.003646	0.004080	7.981	14.394

Wave 3.2 Harmonic-Offset Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2f_bis_harmonic_residual_offset_Bw	0.003331	0.003671	7.261	14.149
track2f_bis_clean_sequential_residual_offset_Bw	0.003541	0.003971	7.732	13.916
track2f_bis_clean_sequential_residual_offset_global	0.003624	0.004058	7.918	12.339
track2f_bis_harmonic_residual_offset_global	0.003805	0.004120	8.354	14.523

Wave 3.3 Curve-Aware Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2g_curve_aware_pointwise_control_Bw	0.003436	0.003761	7.538	13.918
track2g_curve_aware_raw_centered_shape_global	0.003449	0.003808	7.572	14.065
track2g_curve_aware_raw_centered_shape_Bw	0.003465	0.003790	7.582	12.901
track2g_curve_aware_raw_offset_Bw	0.003469	0.003799	7.608	12.513
track2g_curve_aware_full_curve_composite_Bw	0.003510	0.003897	7.683	13.038
track2g_curve_aware_raw_offset_global	0.003559	0.003881	7.831	13.812
track2g_curve_aware_pointwise_control_global	0.003658	0.004016	8.051	15.707
track2g_curve_aware_full_curve_composite_global	0.005829	0.006210	12.816	27.371

Wave 5.2 series Robust-Loss Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_smooth_l1_robust_Bw	0.003078	0.003403	6.676	15.410
track2h_mae_robust_Bw	0.003433	0.003750	7.506	14.575
track2h_log_cosh_robust_global	0.003481	0.003814	7.603	13.596
track2h_log_cosh_robust_Bw	0.003486	0.003811	7.628	12.872
track2h_mae_robust_global	0.003568	0.003886	7.818	14.652
track2h_smooth_l1_robust_global	0.003612	0.003931	7.918	14.825

Wave 5.2 series Quantile Probabilistic Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_quantile_p10_p50_p90_Bw	0.002935	0.003250	6.307	16.529
track2h_gaussian_nll_Bw	0.003001	0.003303	6.488	14.856
track2h_gaussian_nll_global	0.003068	0.003372	6.627	15.928
track2h_quantile_p10_p50_p90_global	0.003563	0.003909	7.816	15.378

Wave 5.2 series Mixture Density Heads Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_mdn_k2_Bw	0.002668	0.002947	5.880	15.593
track2h_mdn_k3_Bw	0.002730	0.003009	6.049	15.512
track2h_mdn_k3_global	0.003701	0.004023	8.129	16.261
track2h_mdn_k2_global	0.003735	0.004073	8.194	15.436

Wave 4.4 Latent-State Hysteresis Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2h_l_causal_tcn_offset_residual_global	0.003478	0.003940	7.555	13.771
track2h_l_gru_offset_residual_Bw	0.003542	0.003984	7.736	12.831
track2h_l_causal_tcn_offset_residual_Bw	0.003624	0.004098	7.903	13.135
track2h_l_gru_offset_residual_global	0.008530	0.008976	18.741	40.030

Wave 5.1 Harmonic-Prior Residual Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
wave3_harmonic_prior_residual_pointwise_control_Bw	0.003360	0.003677	7.363	13.296
wave3_harmonic_prior_residual_smooth_l1_structured_Bw	0.003431	0.003739	7.523	13.200
wave3_harmonic_prior_residual_pointwise_control_global	0.003604	0.003959	7.906	14.071
wave3_harmonic_prior_residual_smooth_l1_structured_global	0.008517	0.008877	18.660	41.964

Wave 5.2B Offset And Harmonic Guided Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
wave52b_offset_centered_shape_harmonic_global	0.004781	0.005129	10.705	26.696
wave52b_offset_centered_shape_harmonic_Bw	0.020690	0.021139	48.162	112.613

Polished Model-Development Backward And Global Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_harmonic_regression_Bw	0.003678	0.004012	8.058	15.071
polished_wave4_2_quantile_p10_p50_p90_Bw	0.003809	0.004175	8.347	17.460
polished_harmonic_regression_global	0.004331	0.004726	9.481	21.909
polished_wave4_1_mae_robust_loss_global	0.004734	0.005066	10.371	22.678
polished_wave4_3_mixture_density_k2_Bw	0.004718	0.005044	10.385	22.069
polished_wave4_2_gaussian_nll_global	0.004723	0.005058	10.446	22.222
polished_wave4_1_log_cosh_robust_loss_Bw	0.004849	0.005168	10.664	25.709
polished_wave4_3_mixture_density_k2_global	0.004832	0.005146	10.763	23.393
polished_wave3_2_harmonic_residual_offset_Bw	0.004950	0.005267	11.001	24.381
polished_wave3_3_raw_offset_curve_aware_global	0.004985	0.005314	11.054	23.547
polished_wave4_3_mixture_density_k3_Bw	0.005049	0.005359	11.212	23.958
polished_wave4_2_gaussian_nll_Bw	0.005195	0.005557	11.506	25.252
polished_wave3_2_harmonic_residual_offset_global	0.005272	0.005623	11.611	25.054
polished_wave3_3_raw_centered_shape_curve_aware_Bw	0.005377	0.005696	11.898	24.443
polished_wave4_2_quantile_p10_p50_p90_global	0.005455	0.005807	11.933	25.004
polished_wave3_3_full_curve_composite_Bw	0.005651	0.006040	12.391	25.915
polished_wave3_3_full_curve_composite_global	0.005763	0.006116	12.644	28.205
polished_wave3_3_curve_aware_pointwise_control_global	0.005754	0.006094	12.647	25.192
polished_wave4_1_smooth_l1_robust_loss_global	0.005958	0.006297	13.003	32.185
polished_wave3_3_raw_centered_shape_curve_aware_global	0.005942	0.006277	13.131	29.183
polished_periodic_gru_sequence_Bw	0.006320	0.006795	14.078	28.125
polished_temporal_convolution_global	0.006542	0.007000	14.321	32.961
polished_wave5_1_harmonic_prior_pointwise_control_global	0.006599	0.006949	14.435	33.050
polished_periodic_temporal_convolution_global	0.006842	0.007285	14.811	35.825
polished_residual_harmonic_gru_sequence_sparse_rcim_global	0.006776	0.007157	14.838	30.081
polished_residual_harmonic_lstm_sequence_dense240_Bw	0.006772	0.007632	14.902	31.588
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	0.006762	0.007113	14.904	33.497
polished_gru_sequence_global	0.006814	0.007285	14.950	30.038
polished_lstm_sequence_Bw	0.006945	0.007367	15.245	39.006
polished_wave5_1_harmonic_prior_smooth_l1_structured_Bw	0.006945	0.007295	15.256	34.459
polished_residual_harmonic_gru_sequence_sparse_rcim_Bw	0.007012	0.007383	15.369	29.992
polished_wave3_3_curve_aware_pointwise_control_Bw	0.007181	0.007525	15.776	33.540
polished_wave4_1_smooth_l1_robust_loss_Bw	0.007174	0.007502	15.817	33.663
polished_wave4_1_log_cosh_robust_loss_global	0.007318	0.007648	16.063	36.570
polished_periodic_gru_sequence_global	0.007336	0.007715	16.264	32.319
polished_lstm_sequence_global	0.007449	0.007853	16.308	41.180

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_gru_sequence_Bw	0.007468	0.007885	16.367	34.142
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	0.007575	0.007936	16.542	44.368
polished_wave5_1_harmonic_prior_pointwise_control_Bw	0.007570	0.007930	16.593	32.876
polished_periodic_temporal_convolution_Bw	0.007596	0.008027	16.602	37.232
polished_residual_harmonic_lstm_sequence_sparse_rcim_Bw	0.007740	0.008099	16.813	44.341
polished_wave4_4_causal_tcn_latent_offset_residual_Bw	0.007758	0.008220	17.080	36.319
polished_wave3_3_raw_offset_curve_aware_Bw	0.007815	0.008139	17.245	38.577
polished_wave4_4_gru_latent_offset_residual_Bw	0.008062	0.008539	17.742	40.408
polished_wave4_1_mae_robust_loss_Bw	0.008153	0.008484	17.810	41.953
polished_feedforward_global	0.008231	0.008800	17.869	42.849
polished_wave3_2_clean_sequential_residual_offset_Bw	0.008107	0.008493	17.879	39.480
polished_feedforward_Bw	0.008079	0.008672	17.892	35.715
polished_wave3_2_clean_sequential_residual_offset_global	0.008207	0.008634	18.039	41.136
polished_residual_harmonic_lstm_sequence_dense360_global	0.008233	0.009936	18.219	37.015
polished_wave4_3_mixture_density_k3_global	0.008362	0.008683	18.302	42.032
polished_wave4_4_gru_latent_offset_residual_global	0.008451	0.008942	18.421	39.768
polished_residual_harmonic_gru_sequence_dense360_global	0.008335	0.010156	18.494	35.068
polished_residual_harmonic_lstm_sequence_dense240_global	0.008410	0.009192	18.525	43.908
polished_residual_harmonic_mlp_Bw	0.008619	0.009001	18.775	45.819
polished_residual_harmonic_mlp_global	0.008685	0.009067	18.784	45.907
polished_wave4_4_causal_tcn_latent_offset_residual_global	0.008738	0.009177	18.907	46.002
polished_wave3_1_sequential_residual_offset_probe_global	0.008771	0.009248	19.185	44.010
polished_temporal_convolution_Bw	0.008956	0.009381	19.524	43.693
polished_periodic_mlp_harmonic_global	0.009199	0.009571	19.896	48.481
polished_residual_harmonic_lstm_sequence_dense360_Bw	0.009108	0.011272	20.181	44.876
polished_periodic_mlp_harmonic_Bw	0.009393	0.009831	20.198	50.399
polished_residual_harmonic_gru_sequence_dense240_Bw	0.009235	0.009959	20.204	42.882
polished_residual_harmonic_gru_sequence_dense240_global	0.009240	0.009932	20.241	44.364
polished_periodic_lstm_sequence_global	0.009636	0.010017	20.812	46.697
polished_periodic_mlp_global	0.009681	0.010115	20.915	51.053
polished_wave3_1_sequential_residual_offset_probe_Bw	0.009712	0.010105	21.306	49.014
polished_periodic_mlp_Bw	0.010754	0.011174	23.160	55.992
polished_residual_harmonic_gru_sequence_dense360_Bw	0.010578	0.012666	23.297	45.514
polished_periodic_lstm_sequence_Bw	0.010769	0.011111	23.306	52.305
polished_tree_global	0.015215	0.015523	34.609	61.506
polished_tree_Bw	0.015215	0.015523	34.609	61.506

Polished RCIM Model-Bank Reproduction Backward Models

Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_rcim_model_bank_reproduction_ERT19_Bw	0.006784	0.007076	15.336	50.015
polished_rcim_model_bank_reproduction_DT19_Bw	0.006786	0.007054	15.367	50.818
polished_rcim_model_bank_reproduction_LGBM19_Bw	0.007018	0.007321	15.907	50.821
polished_rcim_model_bank_reproduction_HGBM19_Bw	0.006968	0.007235	15.935	54.187
polished_rcim_model_bank_reproduction_ET19_Bw	0.007426	0.007704	16.960	62.118
polished_rcim_model_bank_reproduction_GBM19_Bw	0.007624	0.007918	17.434	48.611
polished_rcim_model_bank_reproduction_XGBM19_Bw	0.008480	0.008772	19.156	46.773
polished_rcim_model_bank_reproduction_RF19_Bw	0.009117	0.009403	20.804	52.264
polished_rcim_model_bank_reproduction_SVM19_Bw	0.019584	0.021532	44.418	84.739
polished_rcim_model_bank_reproduction_MLP19_Bw	0.851212	1.066855	1971.861	4215.375

Global Model Direction Breakdown

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
feedforward_global	forward	0.003550	0.003979	7.916	16.195
feedforward_global	backward	0.003741	0.004188	8.186	15.259
feedforward_global	combined	0.003646	0.004083	8.051	15.583
gru_sequence_global	forward	0.003546	0.003975	7.869	14.344
gru_sequence_global	backward	0.003637	0.004080	7.946	14.580
gru_sequence_global	combined	0.003591	0.004028	7.907	14.389
harmonic_regression_global	forward	0.018314	0.018547	41.749	79.404
harmonic_regression_global	backward	0.018006	0.018276	41.275	83.603
harmonic_regression_global	combined	0.018160	0.018412	41.512	82.048
lstm_sequence_global	forward	0.003445	0.003863	7.642	12.032
lstm_sequence_global	backward	0.003515	0.003944	7.666	12.502
lstm_sequence_global	combined	0.003480	0.003903	7.654	12.430
periodic_gru_sequence_global	forward	0.002777	0.003023	6.267	13.580
periodic_gru_sequence_global	backward	0.002630	0.002876	6.010	12.693
periodic_gru_sequence_global	combined	0.002704	0.002949	6.139	13.200
periodic_lstm_sequence_global	forward	0.002726	0.002959	6.142	14.092
periodic_lstm_sequence_global	backward	0.002689	0.002956	6.098	14.674
periodic_lstm_sequence_global	combined	0.002707	0.002958	6.120	14.717
periodic_mlp_global	forward	0.003380	0.003791	7.493	14.349
periodic_mlp_global	backward	0.003610	0.004030	7.879	13.661
periodic_mlp_global	combined	0.003495	0.003910	7.686	14.284
periodic_temporal_convolution_global	forward	0.003407	0.003724	7.581	14.480
periodic_temporal_convolution_global	backward	0.003604	0.003948	7.935	13.804
periodic_temporal_convolution_global	combined	0.003506	0.003836	7.758	14.308
polished_feedforward_global	backward	0.008231	0.008800	17.869	42.849
polished_feedforward_global	combined	0.008231	0.008800	17.869	42.849

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_gru_sequence_global	backward	0.006814	0.007285	14.950	30.038
polished_gru_sequence_global	combined	0.006814	0.007285	14.950	30.038
polished_harmonic_regression_global	backward	0.004331	0.004726	9.481	21.909
polished_harmonic_regression_global	combined	0.004331	0.004726	9.481	21.909
polished_lstm_sequence_global	backward	0.007449	0.007853	16.308	41.180
polished_lstm_sequence_global	combined	0.007449	0.007853	16.308	41.180
polished_periodic_gru_sequence_global	backward	0.007336	0.007715	16.264	32.319
polished_periodic_gru_sequence_global	combined	0.007336	0.007715	16.264	32.319
polished_periodic_lstm_sequence_global	backward	0.009636	0.010017	20.812	46.697
polished_periodic_lstm_sequence_global	combined	0.009636	0.010017	20.812	46.697
polished_periodic_mlp_global	backward	0.009681	0.010115	20.915	51.053
polished_periodic_mlp_global	combined	0.009681	0.010115	20.915	51.053
polished_periodic_mlp_harmonic_global	backward	0.009199	0.009571	19.896	48.481
polished_periodic_mlp_harmonic_global	combined	0.009199	0.009571	19.896	48.481
polished_periodic_temporal_convolution_global	backward	0.006842	0.007285	14.811	35.825
polished_periodic_temporal_convolution_global	combined	0.006842	0.007285	14.811	35.825
polished_residual_harmonic_gru_sequence_dense240_global	backward	0.009240	0.009932	20.241	44.364
polished_residual_harmonic_gru_sequence_dense240_global	combined	0.009240	0.009932	20.241	44.364
polished_residual_harmonic_gru_sequence_dense360_global	backward	0.008335	0.010156	18.494	35.068
polished_residual_harmonic_gru_sequence_dense360_global	combined	0.008335	0.010156	18.494	35.068
polished_residual_harmonic_gru_sequence_sparse_rcim_global	backward	0.006776	0.007157	14.838	30.081
polished_residual_harmonic_gru_sequence_sparse_rcim_global	combined	0.006776	0.007157	14.838	30.081
polished_residual_harmonic_lstm_sequence_dense240_global	backward	0.008410	0.009192	18.525	43.908
polished_residual_harmonic_lstm_sequence_dense240_global	combined	0.008410	0.009192	18.525	43.908
polished_residual_harmonic_lstm_sequence_dense360_global	backward	0.008233	0.009936	18.219	37.015
polished_residual_harmonic_lstm_sequence_dense360_global	combined	0.008233	0.009936	18.219	37.015
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	backward	0.007575	0.007936	16.542	44.368
polished_residual_harmonic_lstm_sequence_sparse_rcim_global	combined	0.007575	0.007936	16.542	44.368
polished_residual_harmonic_mlp_global	backward	0.008685	0.009067	18.784	45.907
polished_residual_harmonic_mlp_global	combined	0.008685	0.009067	18.784	45.907
polished_temporal_convolution_global	backward	0.006542	0.007000	14.321	32.961
polished_temporal_convolution_global	combined	0.006542	0.007000	14.321	32.961
polished_tree_global	backward	0.015215	0.015523	34.609	61.506
polished_tree_global	combined	0.015215	0.015523	34.609	61.506
polished_wave3_1_sequential_residual_offset_probe_global	backward	0.008771	0.009248	19.185	44.010
polished_wave3_1_sequential_residual_offset_probe_global	combined	0.008771	0.009248	19.185	44.010

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_wave3_2_clean_sequential_residual_offset_global	backward	0.008207	0.008634	18.039	41.136
polished_wave3_2_clean_sequential_residual_offset_global	combined	0.008207	0.008634	18.039	41.136
polished_wave3_2_harmonic_residual_offset_global	backward	0.005272	0.005623	11.611	25.054
polished_wave3_2_harmonic_residual_offset_global	combined	0.005272	0.005623	11.611	25.054
polished_wave3_3_curve_aware_pointwise_control_global	backward	0.005754	0.006094	12.647	25.192
polished_wave3_3_curve_aware_pointwise_control_global	combined	0.005754	0.006094	12.647	25.192
polished_wave3_3_full_curve_composite_global	backward	0.005763	0.006116	12.644	28.205
polished_wave3_3_full_curve_composite_global	combined	0.005763	0.006116	12.644	28.205
polished_wave3_3_raw_centered_shape_curve_aware_global	backward	0.005942	0.006277	13.131	29.183
polished_wave3_3_raw_centered_shape_curve_aware_global	combined	0.005942	0.006277	13.131	29.183
polished_wave3_3_raw_offset_curve_aware_global	backward	0.004985	0.005314	11.054	23.547
polished_wave3_3_raw_offset_curve_aware_global	combined	0.004985	0.005314	11.054	23.547
polished_wave4_1_log_cosh_robust_loss_global	backward	0.007318	0.007648	16.063	36.570
polished_wave4_1_log_cosh_robust_loss_global	combined	0.007318	0.007648	16.063	36.570
polished_wave4_1_mae_robust_loss_global	backward	0.004734	0.005066	10.371	22.678
polished_wave4_1_mae_robust_loss_global	combined	0.004734	0.005066	10.371	22.678
polished_wave4_1_smooth_l1_robust_loss_global	backward	0.005958	0.006297	13.003	32.185
polished_wave4_1_smooth_l1_robust_loss_global	combined	0.005958	0.006297	13.003	32.185
polished_wave4_2_gaussian_nll_global	backward	0.004723	0.005058	10.446	22.222
polished_wave4_2_gaussian_nll_global	combined	0.004723	0.005058	10.446	22.222
polished_wave4_2_quantile_p10_p50_p90_global	backward	0.005455	0.005807	11.933	25.004
polished_wave4_2_quantile_p10_p50_p90_global	combined	0.005455	0.005807	11.933	25.004
polished_wave4_3_mixture_density_k2_global	backward	0.004832	0.005146	10.763	23.393
polished_wave4_3_mixture_density_k2_global	combined	0.004832	0.005146	10.763	23.393
polished_wave4_3_mixture_density_k3_global	backward	0.008362	0.008683	18.302	42.032
polished_wave4_3_mixture_density_k3_global	combined	0.008362	0.008683	18.302	42.032
polished_wave4_4_causal_tcn_latent_offset_residual_global	backward	0.008738	0.009177	18.907	46.002
polished_wave4_4_causal_tcn_latent_offset_residual_global	combined	0.008738	0.009177	18.907	46.002
polished_wave4_4_gru_latent_offset_residual_global	backward	0.008451	0.008942	18.421	39.768
polished_wave4_4_gru_latent_offset_residual_global	combined	0.008451	0.008942	18.421	39.768
polished_wave5_1_harmonic_prior_pointwise_control_global	backward	0.006599	0.006949	14.435	33.050
polished_wave5_1_harmonic_prior_pointwise_control_global	combined	0.006599	0.006949	14.435	33.050
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	backward	0.006762	0.007113	14.904	33.497

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
polished_wave5_1_harmonic_prior_smooth_l1_structured_global	combined	0.006762	0.007113	14.904	33.497
residual_harmonic_gru_sequence_dense240_global	forward	0.006534	0.008959	14.732	20.898
residual_harmonic_gru_sequence_dense240_global	backward	0.006786	0.009221	15.282	20.990
residual_harmonic_gru_sequence_dense240_global	combined	0.006660	0.009090	15.007	20.940
residual_harmonic_gru_sequence_dense360_global	forward	0.007844	0.011211	17.679	22.520
residual_harmonic_gru_sequence_dense360_global	backward	0.008181	0.011621	18.502	23.968
residual_harmonic_gru_sequence_dense360_global	combined	0.008012	0.011416	18.090	23.559
residual_harmonic_gru_sequence_sparse_rcim_global	forward	0.003287	0.003611	7.282	12.461
residual_harmonic_gru_sequence_sparse_rcim_global	backward	0.003584	0.003957	7.859	14.946
residual_harmonic_gru_sequence_sparse_rcim_global	combined	0.003435	0.003784	7.571	13.869
residual_harmonic_lstm_sequence_dense240_global	forward	0.006247	0.008597	14.061	20.066
residual_harmonic_lstm_sequence_dense240_global	backward	0.006591	0.008932	14.858	21.728
residual_harmonic_lstm_sequence_dense240_global	combined	0.006419	0.008765	14.460	20.628
residual_harmonic_lstm_sequence_dense360_global	forward	0.008760	0.012992	19.759	24.068
residual_harmonic_lstm_sequence_dense360_global	backward	0.008860	0.013061	20.073	23.940
residual_harmonic_lstm_sequence_dense360_global	combined	0.008810	0.013026	19.916	24.052
residual_harmonic_lstm_sequence_sparse_rcim_global	forward	0.003245	0.003562	7.192	12.555
residual_harmonic_lstm_sequence_sparse_rcim_global	backward	0.003491	0.003876	7.627	13.540
residual_harmonic_lstm_sequence_sparse_rcim_global	combined	0.003368	0.003719	7.409	13.041
residual_harmonic_mlp_global	forward	0.003649	0.004051	8.123	14.713
residual_harmonic_mlp_global	backward	0.003639	0.004044	7.967	14.489
residual_harmonic_mlp_global	combined	0.003644	0.004048	8.045	14.717
sequential_residual_offset_probe_global	backward	0.003646	0.004080	7.981	14.394
sequential_residual_offset_probe_global	combined	0.003646	0.004080	7.981	14.394
temporal_convolution_global	forward	0.003508	0.003928	7.792	14.045
temporal_convolution_global	backward	0.003994	0.004438	8.798	18.339
temporal_convolution_global	combined	0.003751	0.004183	8.295	17.247
track2f_bis_clean_sequential_residual_offset_global	backward	0.003624	0.004058	7.918	12.339
track2f_bis_clean_sequential_residual_offset_global	combined	0.003624	0.004058	7.918	12.339
track2f_bis_harmonic_residual_offset_global	backward	0.003805	0.004120	8.354	14.523
track2f_bis_harmonic_residual_offset_global	combined	0.003805	0.004120	8.354	14.523
track2g_curve_aware_full_curve_composite_global	backward	0.005829	0.006210	12.816	27.371
track2g_curve_aware_full_curve_composite_global	combined	0.005829	0.006210	12.816	27.371
track2g_curve_aware_pointwise_control_global	backward	0.003658	0.004016	8.051	15.707
track2g_curve_aware_pointwise_control_global	combined	0.003658	0.004016	8.051	15.707

Candidate	Direction	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
track2g_curve_aware_raw_centered_shape_global	backward	0.003449	0.003808	7.572	14.065
track2g_curve_aware_raw_centered_shape_global	combined	0.003449	0.003808	7.572	14.065
track2g_curve_aware_raw_offset_global	backward	0.003559	0.003881	7.831	13.812
track2g_curve_aware_raw_offset_global	combined	0.003559	0.003881	7.831	13.812
track2h_gaussian_nll_global	backward	0.003068	0.003372	6.627	15.928
track2h_gaussian_nll_global	combined	0.003068	0.003372	6.627	15.928
track2h_l_causal_tcn_offset_residual_global	backward	0.003478	0.003940	7.555	13.771
track2h_l_causal_tcn_offset_residual_global	combined	0.003478	0.003940	7.555	13.771
track2h_l_gru_offset_residual_global	backward	0.008530	0.008976	18.741	40.030
track2h_l_gru_offset_residual_global	combined	0.008530	0.008976	18.741	40.030
track2h_log_cosh_robust_global	backward	0.003481	0.003814	7.603	13.596
track2h_log_cosh_robust_global	combined	0.003481	0.003814	7.603	13.596
track2h_mae_robust_global	backward	0.003568	0.003886	7.818	14.652
track2h_mae_robust_global	combined	0.003568	0.003886	7.818	14.652
track2h_mdn_k2_global	backward	0.003735	0.004073	8.194	15.436
track2h_mdn_k2_global	combined	0.003735	0.004073	8.194	15.436
track2h_mdn_k3_global	backward	0.003701	0.004023	8.129	16.261
track2h_mdn_k3_global	combined	0.003701	0.004023	8.129	16.261
track2h_quantile_p10_p50_p90_global	backward	0.003563	0.003909	7.816	15.378
track2h_quantile_p10_p50_p90_global	combined	0.003563	0.003909	7.816	15.378
track2h_smooth_l1_robust_global	backward	0.003612	0.003931	7.918	14.825
track2h_smooth_l1_robust_global	combined	0.003612	0.003931	7.918	14.825
tree_global	forward	0.002998	0.003364	6.590	11.601
tree_global	backward	0.003290	0.003702	7.118	13.703
tree_global	combined	0.003144	0.003533	6.854	13.314
wave3_harmonic_prior_residual_pointwise_control_global	backward	0.003604	0.003959	7.906	14.071
wave3_harmonic_prior_residual_pointwise_control_global	combined	0.003604	0.003959	7.906	14.071
wave3_harmonic_prior_residual_smooth_l1_structured_global	backward	0.008517	0.008877	18.660	41.964
wave3_harmonic_prior_residual_smooth_l1_structured_global	combined	0.008517	0.008877	18.660	41.964
wave52b_offset_centered_shape_harmonic_global	backward	0.004781	0.005129	10.705	26.696
wave52b_offset_centered_shape_harmonic_global	combined	0.004781	0.005129	10.705	26.696

Artifacts

- summary YAML: `output\validation_checks\track2_reference_comparison\2026-07-05-13-43-55__track2_full_directional_family_matrix_track2_dataset_surface_simplified_dataset_backward_2026_07_04\validation_summary.yaml` ;
- per-condition CSV: `output\validation_checks\track2_reference_comparison\2026-07-05-13-43-55__track2_full_directional_family_matrix_track2_dataset_surface_simplified_dataset_backward_2026_07_04\per_condition_metrics.csv` ;
- grouped report plot root: `doc\reports\campaign_results\track_2\verification_plots` ;
- grouped report plot count: `0` ;

Interpretation

Rows are ranked by mean percentage error within each source group and direction. Directional paper-reference, Wave 1, and Wave 2.1 models are never evaluated on the opposite direction. Global Wave models remain valid on both directions and are therefore shown in the directional sections and again in the global breakdown. The `rcim_track1` forward reference banks use the opposite stored `h0` sign convention relative to the TE Curve Verification Pipeline reconstruction contract, so the TE Curve Verification Pipeline comparison applies the documented source-specific `h0` compatibility multiplier before curve reconstruction.

Open Gaps

- This remains an offline TE-curve comparison and does not replace the future online `Table 9` compensation benchmark.
- The report uses the saved Python model artifacts from `models/` ; ONNX parity checks remain a separate deployment-readiness task.